TP1 - Exercise 3: Answers

Q5. Why are '-' and 'na' not detected as missing?

For pandas, '-' and 'na' are just text strings, like any other word, so isna() does not count them. Only real empty values (NaN) are counted as missing. Pandas converts only empty cells (and a few default strings when reading a CSV) to NaN, so the other markers must be replaced manually.

Q6. Conversion rule

For each text column, we remove the spaces and put the text in lowercase, then we replace the values '-' and 'na' by np.nan. This way '-', 'NA' or ' na ' are all detected.

Q7. Imputation strategies

For the numerical columns (Salary, Bonus %) we used the median, because it is not sensitive to extreme values. For the categorical columns (Gender, Senior Management, Team...) we used the most frequent value. We did not impute the first name, because it is an identifier and inventing a name has no meaning.

Q8. Limits of imputation

The imputed values are guesses, not real data. Using the median or the most frequent value reduces the variability of the data, can add bias, and does not use the relation between columns. Also, in a real machine learning pipeline, the imputer must be fitted on the training data only and then applied to the test data, to avoid data leakage.